



Leibniz Supercomputing Centre
of the Bavarian Academy of Sciences and Humanities

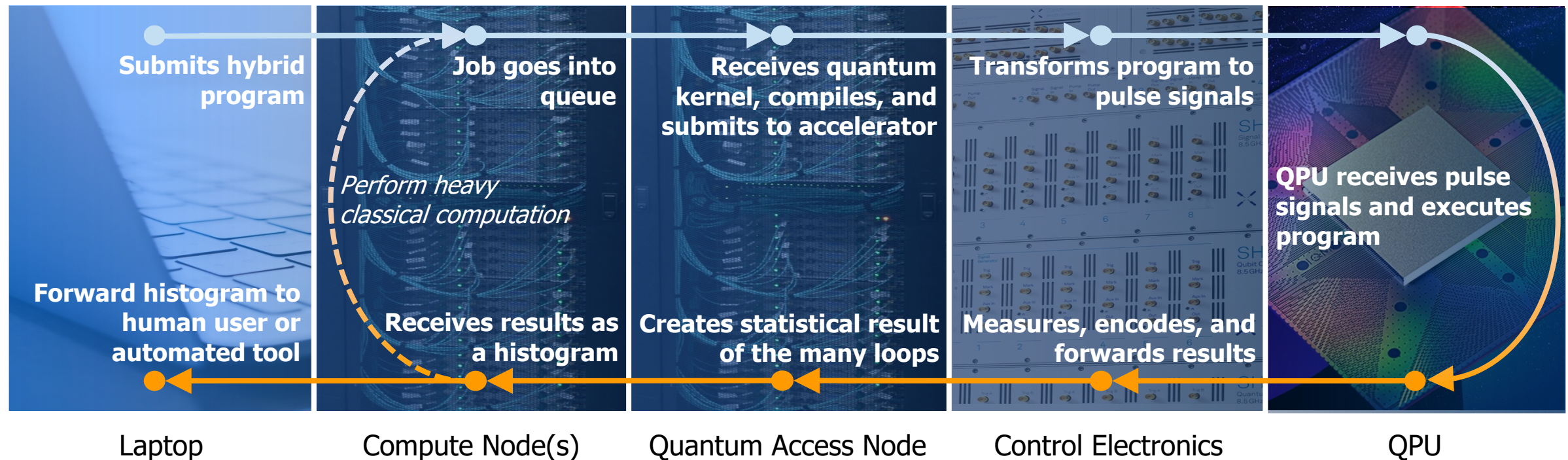
MQSS and HPCQC Integration
Ercüment Kaya - Leibniz Supercomputing Centre
Euro-Q-Exa kick-off
11/03/2026

How do users typically interact with a QC?



In HPC, there is a delicate ballet of resources

We need to operate within the existing framework



Full Quantum Software Stack Problem

Advanced



Intermediate



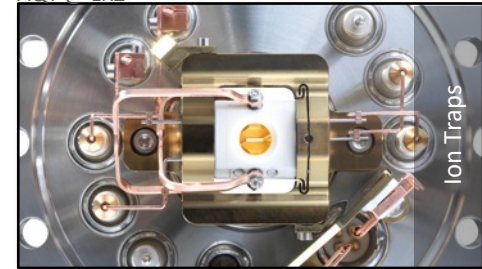
Basic



m types of users

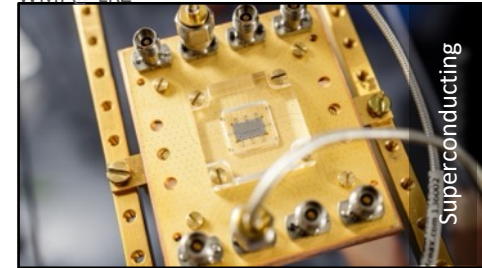
n available devices

AQT @ LRZ



Ion Traps

WMI @ LRZ



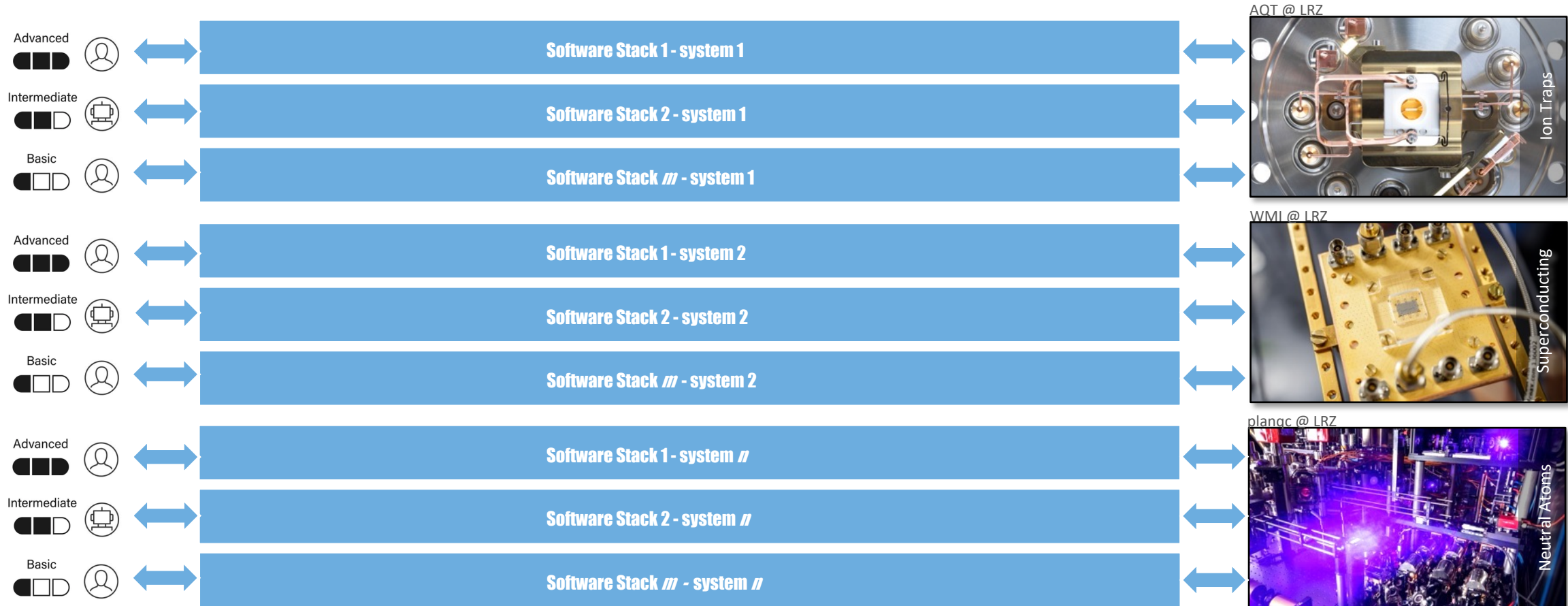
Superconducting

planqc @ LRZ

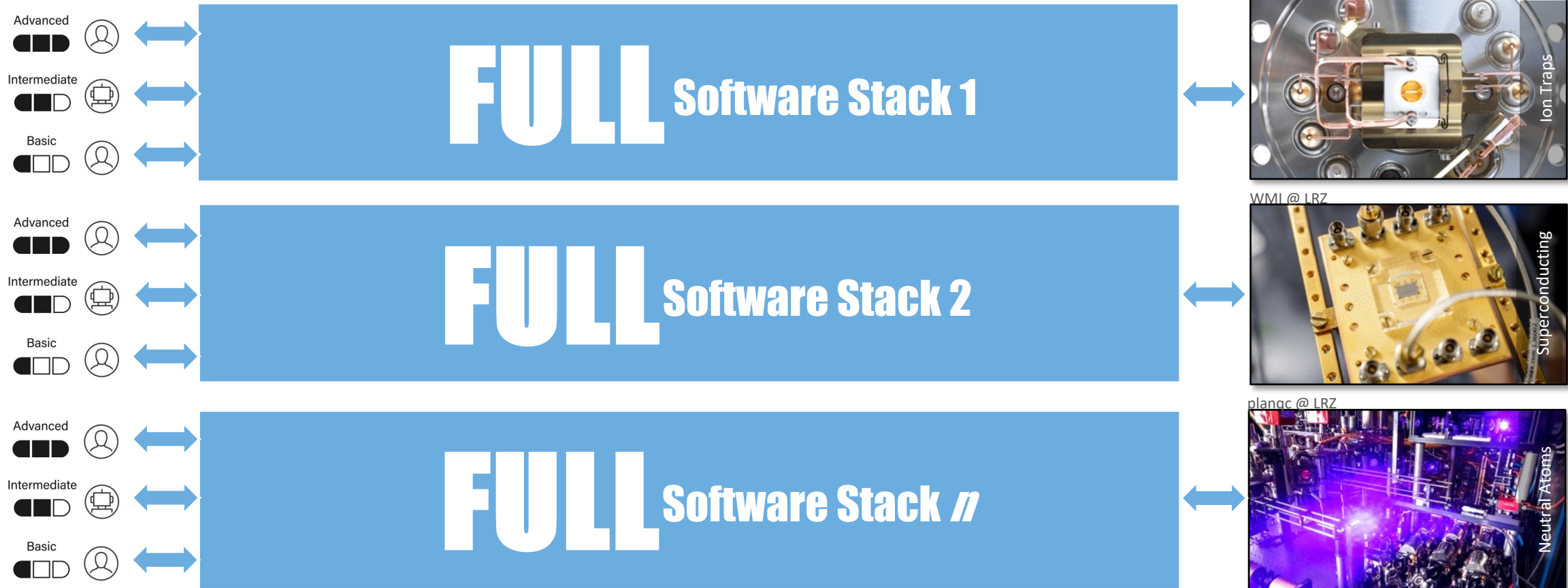


Neutral Atoms

Full Quantum Software Stack Status Quo



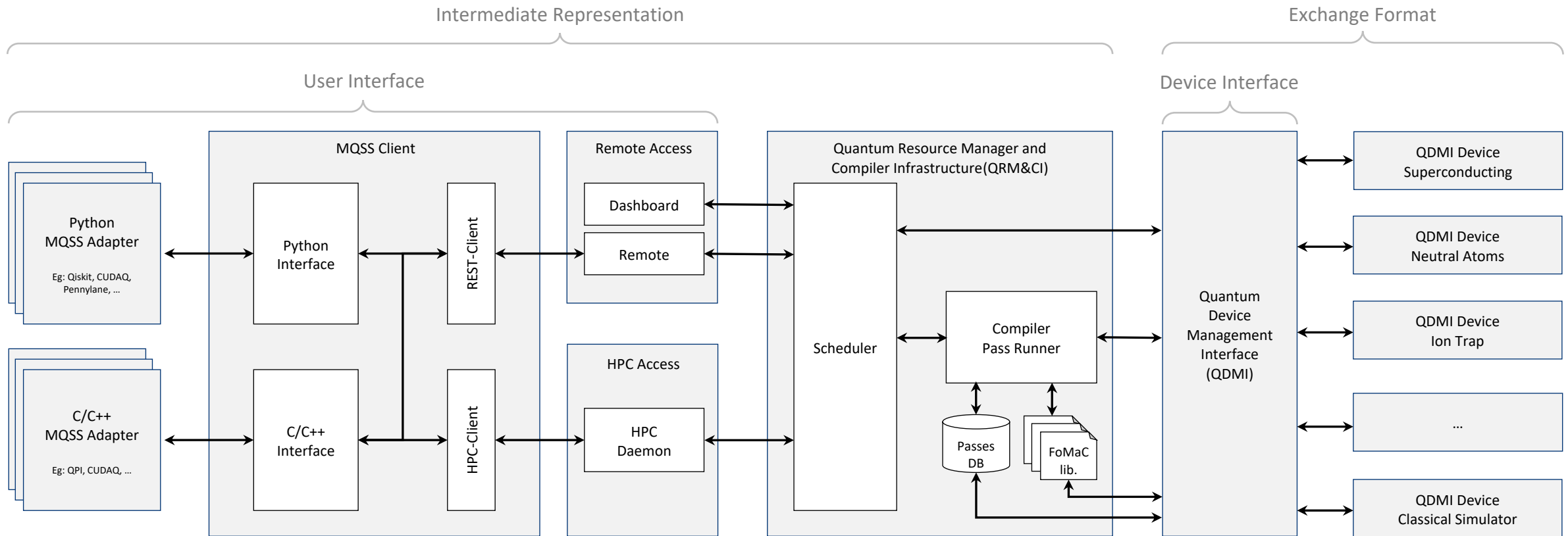
Full Quantum Software Stack Status Quo



A Heterogeneous **Full** Quantum Software Stack

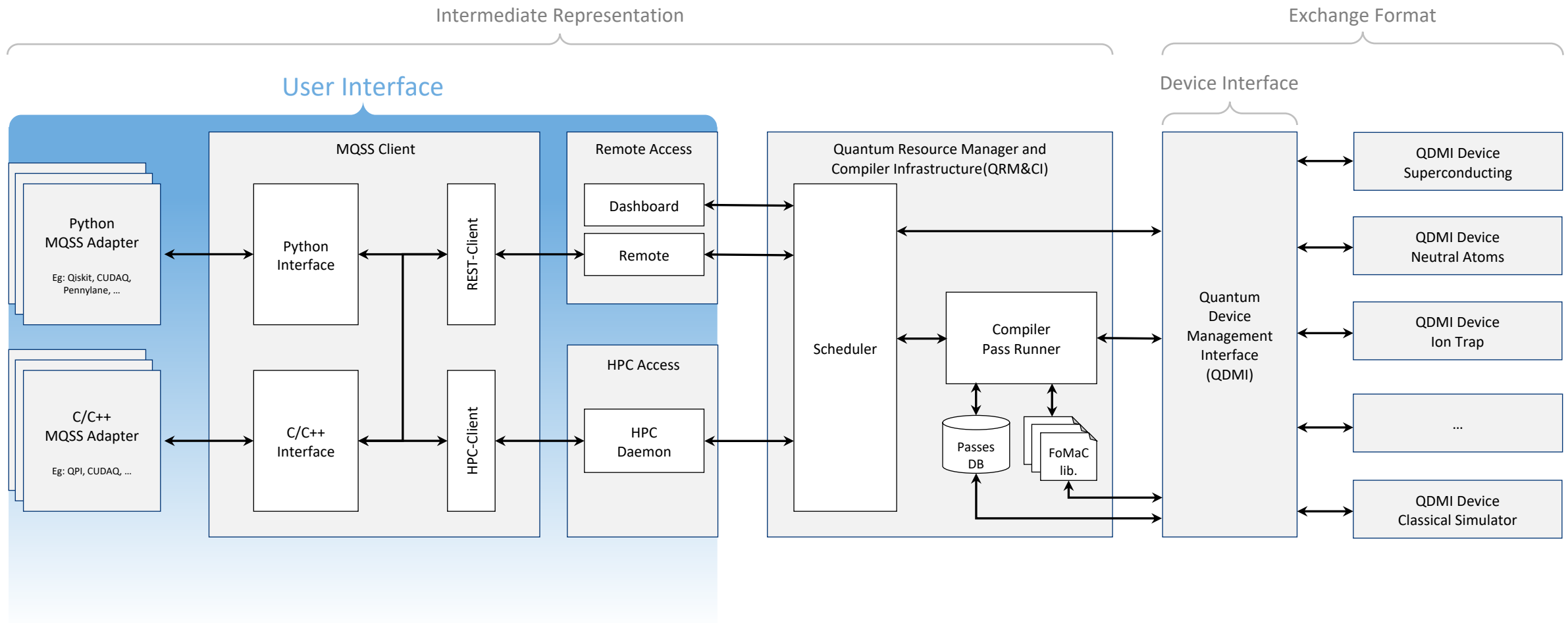


A Heterogeneous **Full** Quantum Software Stack



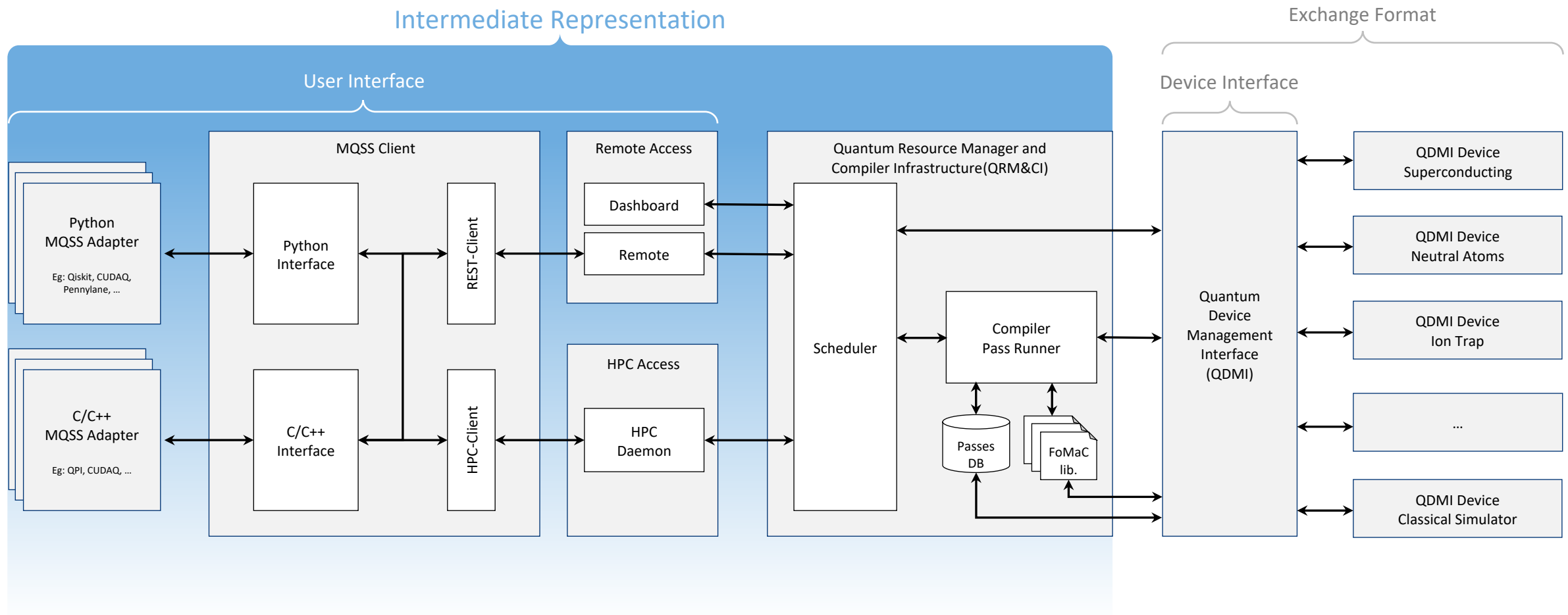
Munich Quantum Software Stack - MQSS

Front-end



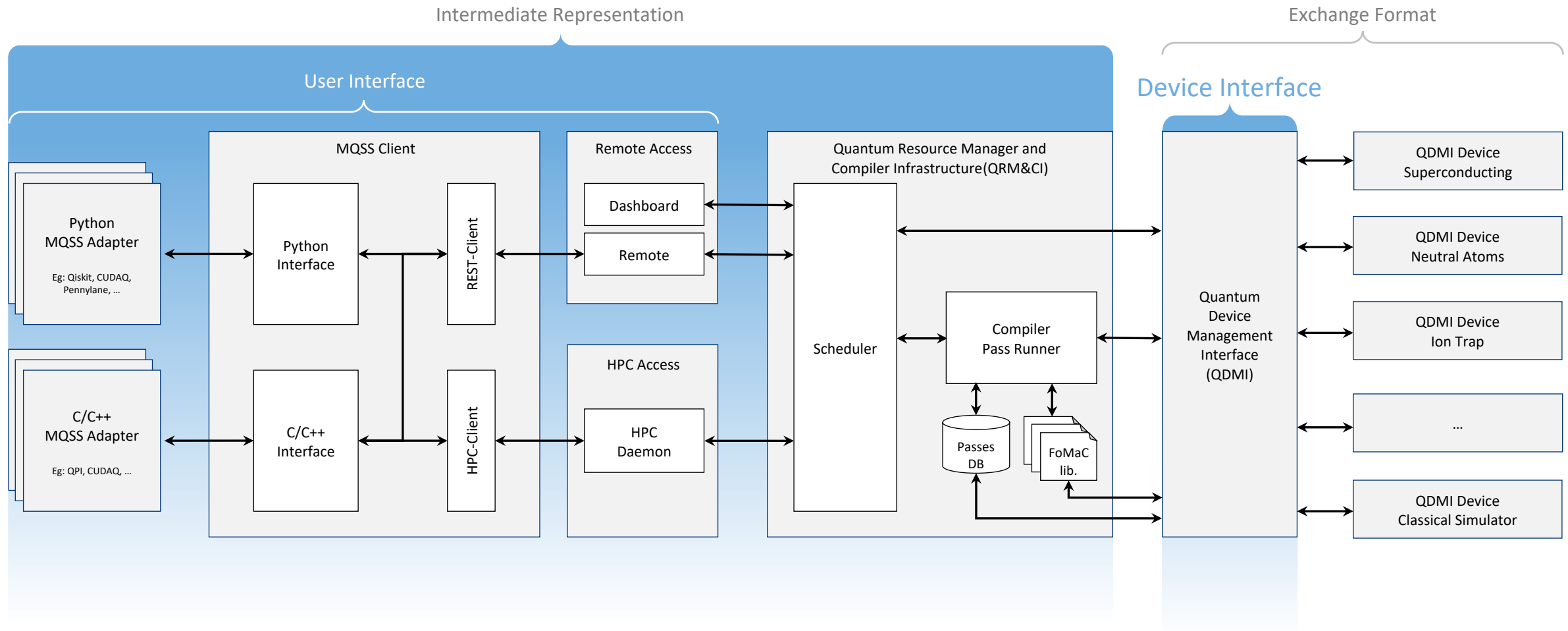
Munich Quantum Software Stack - MQSS

Middle-end



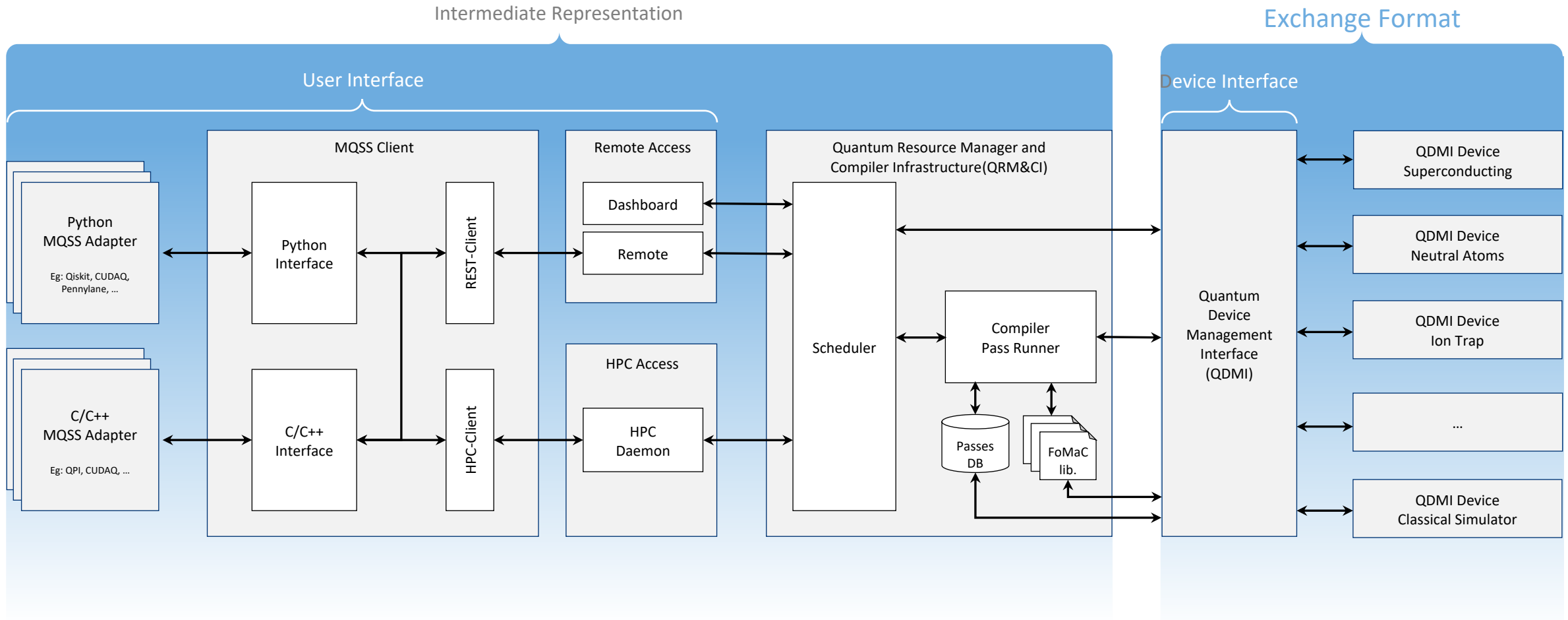
Munich Quantum Software Stack - MQSS

Back-end



Munich Quantum Software Stack - MQSS

Back-end



Questions?